

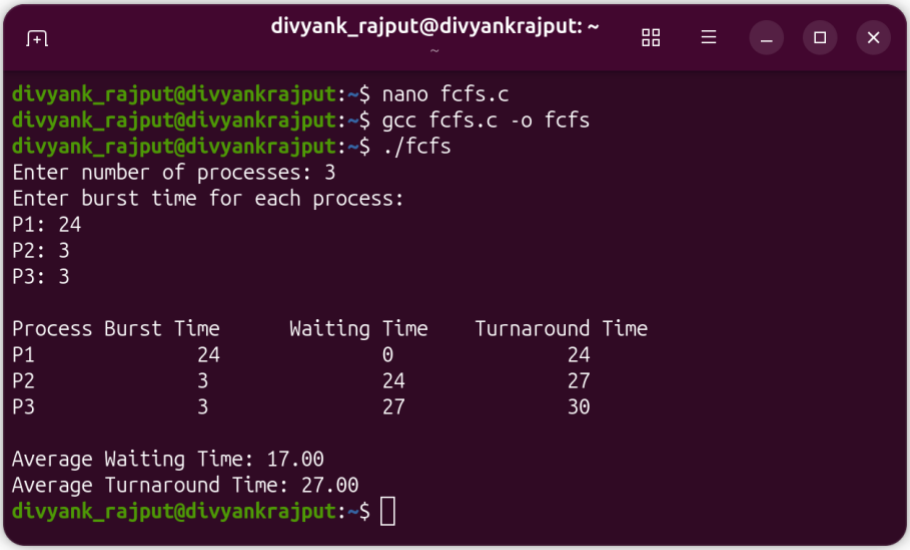
THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Assignment 6:
Operating Systems
(MCA-103)

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2026010038
1MCA-B

Q1. Write a C program to implement the First Come First Serve (FCFS) CPU scheduling algorithm. Calculate the waiting time and turnaround time for each process, along with the average waiting time and average turnaround time.

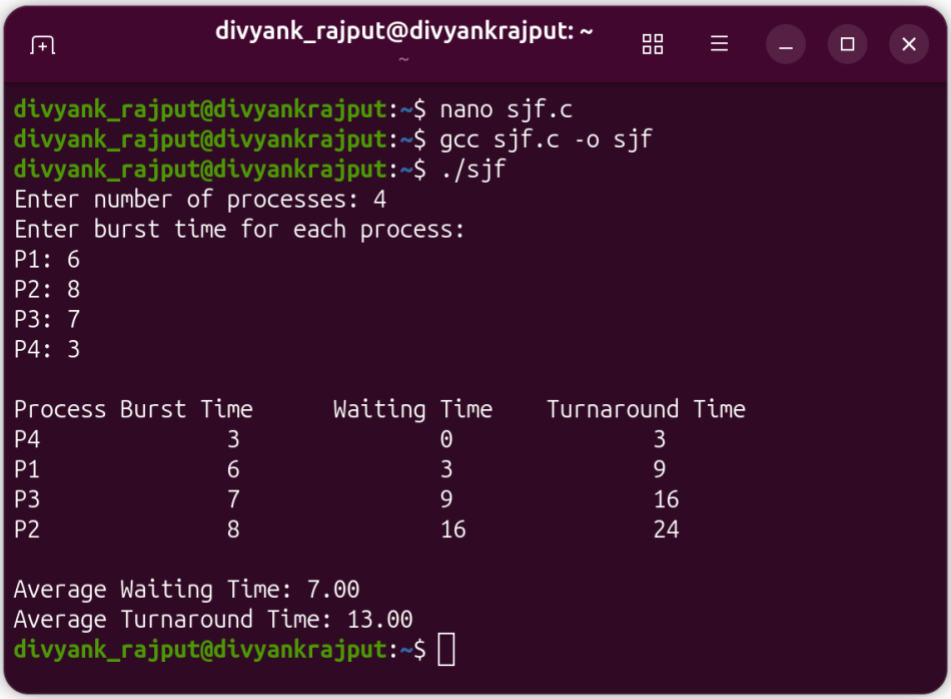




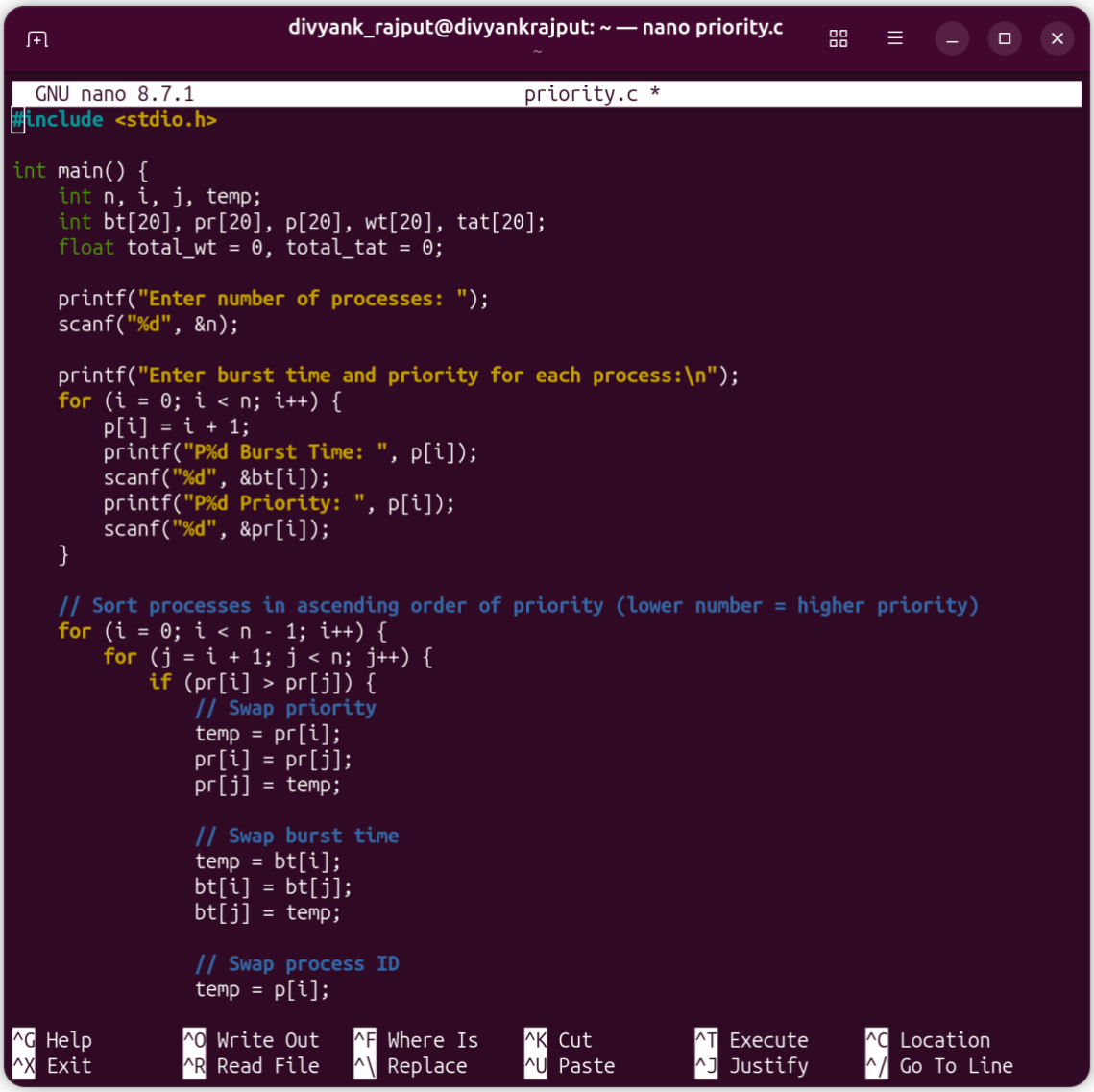
Q2. Write a C program to implement the non-preemptive Shortest Job First (SJF) CPU scheduling algorithm. Calculate the waiting time and turnaround time for each process, along with their respective averages

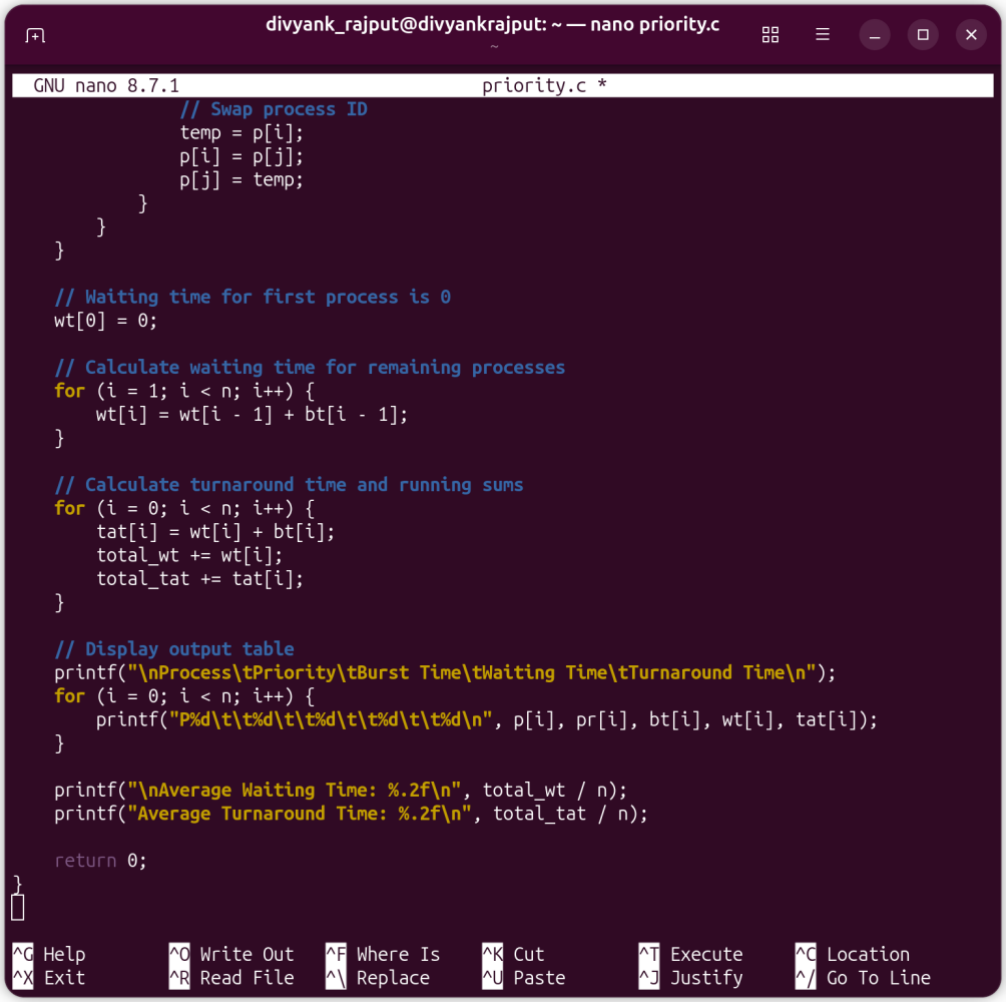


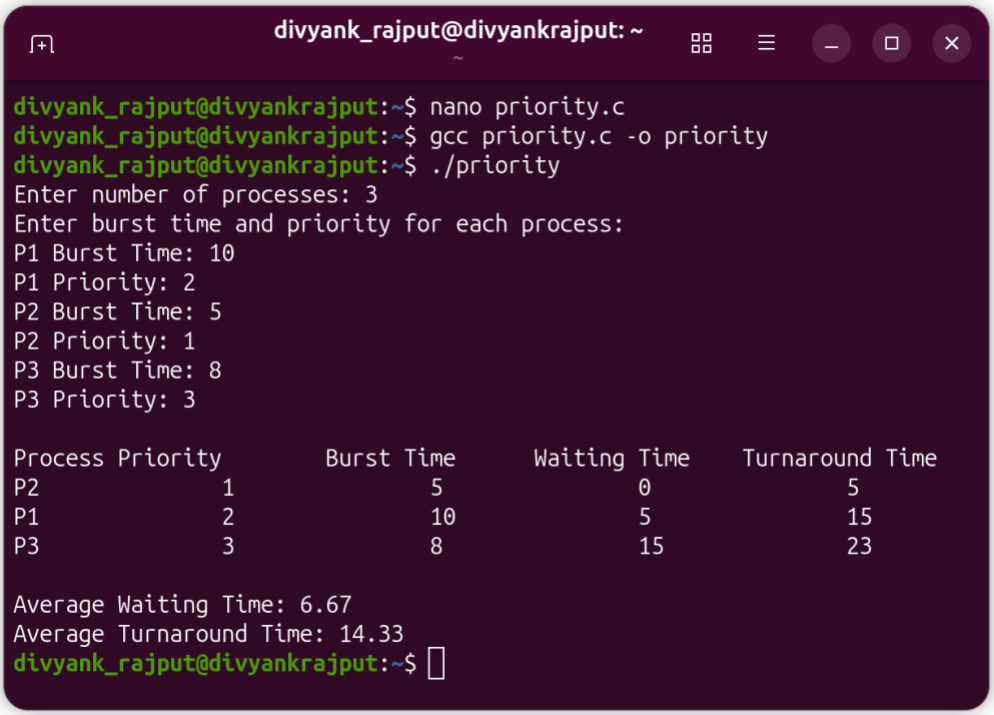

divyank_rajput@divyankrajput: ~ — nano sjf.c
GNU nano 8.7.1 sjf.c *
// Swap process ID
temp = p[i];
p[i] = p[j];
p[j] = temp;
}
}
}
// Waiting time for first process is 0
wt[0] = 0;
// Calculate waiting time for remaining processes
for (i = 1; i < n; i++) {
wt[i] = wt[i - 1] + bt[i - 1];
}
// Calculate turnaround time and accumulators
for (i = 0; i < n; i++) {
tat[i] = wt[i] + bt[i];
total_wt += wt[i];
total_tat += tat[i];
}
// Display output table
printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
for (i = 0; i < n; i++) {
printf("P%d\t\t%d\t\t%d\t\t%d\n", p[i], bt[i], wt[i], tat[i]);
}
printf("\nAverage Waiting Time: %.2f\n", total_wt / n);
printf("Average Turnaround Time: %.2f\n", total_tat / n);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


divyank_rajput@divyankrajput: ~
divyank_rajput@divyankrajput:~$ nano sjf.c
divyank_rajput@divyankrajput:~$ gcc sjf.c -o sjf
divyank_rajput@divyankrajput:~$ ./sjf
Enter number of processes: 4
Enter burst time for each process:
P1: 6
P2: 8
P3: 7
P4: 3
Process Burst Time Waiting Time Turnaround Time
P4 3 0 3
P1 6 3 9
P3 7 9 16
P2 8 16 24
Average Waiting Time: 7.00
Average Turnaround Time: 13.00
divyank_rajput@divyankrajput:~$

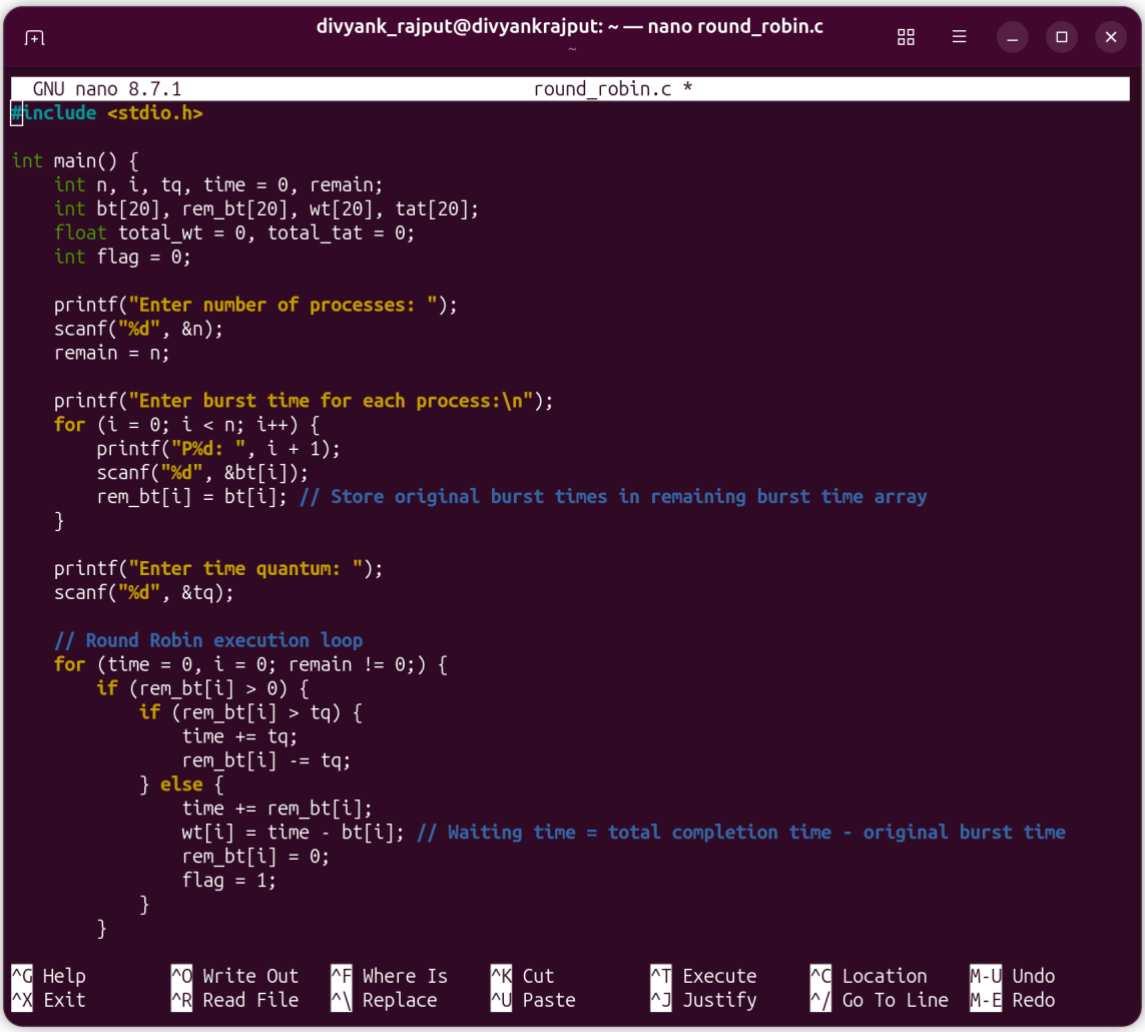
Q3. Write a C program to implement the non-preemptive Priority CPU scheduling algorithm. Accept the burst time and priority of each process.

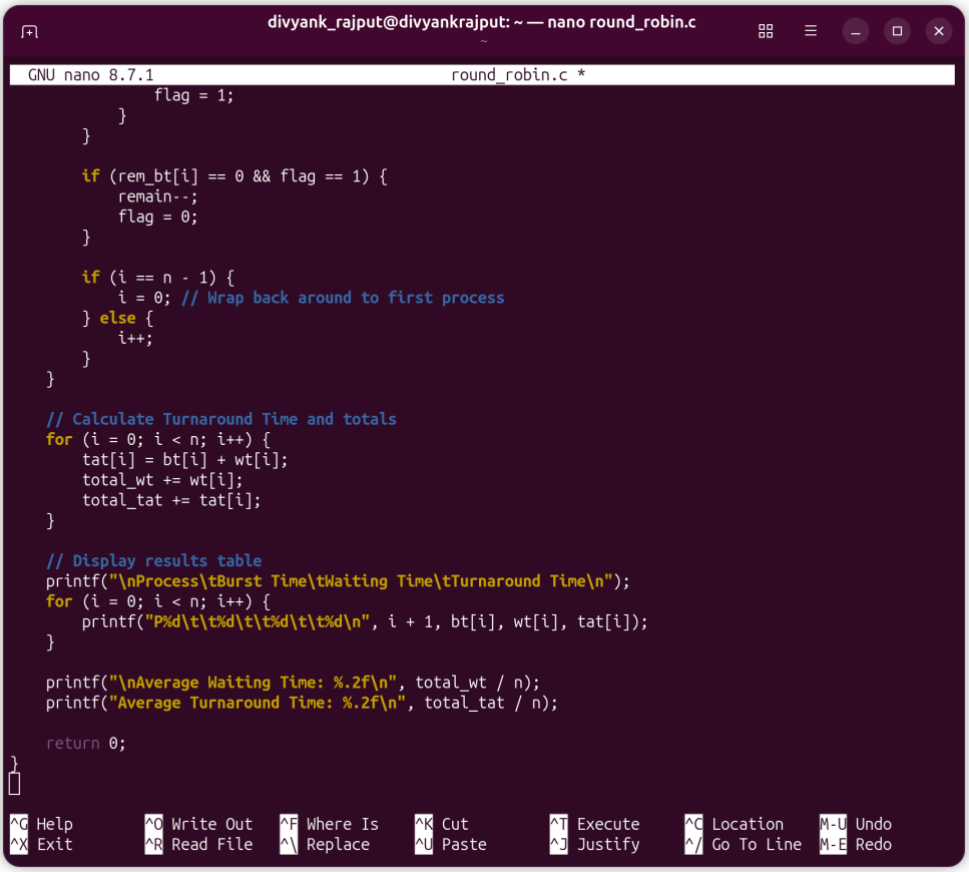


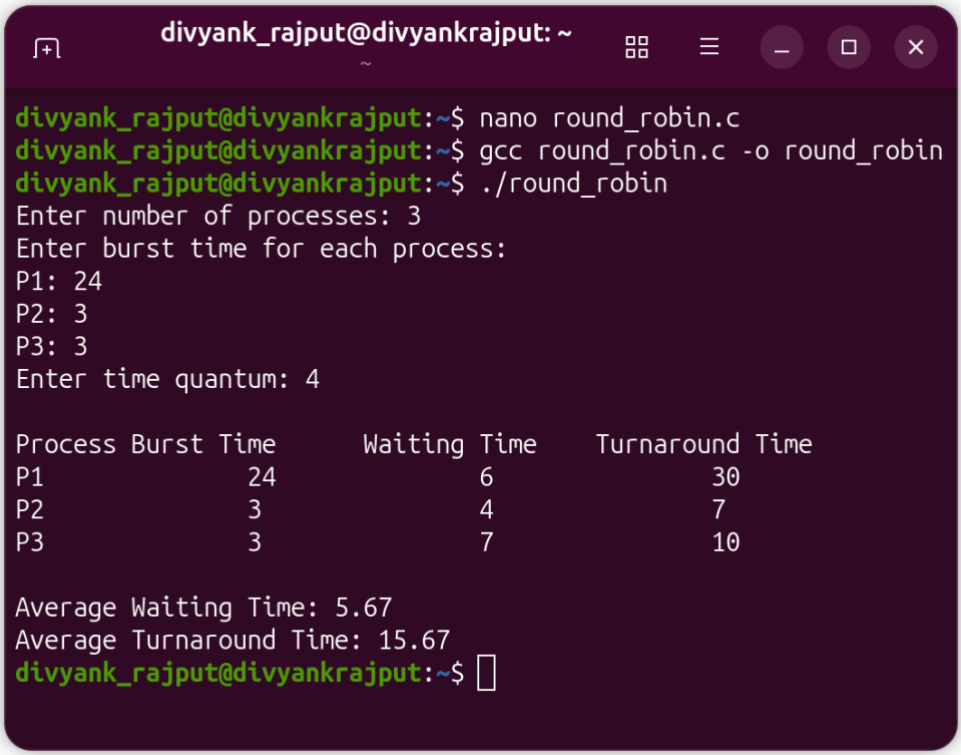

divyank_rajput@divyankrajput: ~ — nano priority.c
GNU nano 8.7.1 priority.c *
// Swap process ID
temp = p[i];
p[i] = p[j];
p[j] = temp;
}
}
}
// Waiting time for first process is 0
wt[0] = 0;
// Calculate waiting time for remaining processes
for (i = 1; i < n; i++) {
wt[i] = wt[i - 1] + bt[i - 1];
}
// Calculate turnaround time and running sums
for (i = 0; i < n; i++) {
tat[i] = wt[i] + bt[i];
total_wt += wt[i];
total_tat += tat[i];
}
// Display output table
printf("\nProcess\tPriority\tBurst Time\tWaiting Time\tTurnaround Time\n");
for (i = 0; i < n; i++) {
printf("P%d\t\t%d\t\t%d\t\t%d\t\t%d\n", p[i], pr[i], bt[i], wt[i], tat[i]);
}
printf("\nAverage Waiting Time: %.2f\n", total_wt / n);
printf("Average Turnaround Time: %.2f\n", total_tat / n);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


divyank_rajput@divyankrajput: ~
divyank_rajput@divyankrajput:~$ nano priority.c
divyank_rajput@divyankrajput:~$ gcc priority.c -o priority
divyank_rajput@divyankrajput:~$ ./priority
Enter number of processes: 3
Enter burst time and priority for each process:
P1 Burst Time: 10
P1 Priority: 2
P2 Burst Time: 5
P2 Priority: 1
P3 Burst Time: 8
P3 Priority: 3
Process Priority Burst Time Waiting Time Turnaround Time
P2 1 5 0 5
P1 2 10 5 15
P3 3 8 15 23
Average Waiting Time: 6.67
Average Turnaround Time: 14.33
divyank_rajput@divyankrajput:~$

Q4. Write a C program to implement the Round Robin CPU scheduling algorithm. Accept the burst time of each process and the time quantum. Calculate the waiting time and turnaround time for each process, along with the average waiting time and average turnaround time.




divyank_rajput@divyankrajput: ~ — nano round_robin.c
GNU nano 8.7.1 round_robin.c *
flag = 1;
}
}
if (rem_bt[i] == 0 && flag == 1) {
remain--;
flag = 0;
}
if (i == n - 1) {
i = 0; // Wrap back around to first process
} else {
i++;
}
}
// Calculate Turnaround Time and totals
for (i = 0; i < n; i++) {
tat[i] = bt[i] + wt[i];
total_wt += wt[i];
total_tat += tat[i];
}
// Display results table
printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
for (i = 0; i < n; i++) {
printf("P%d\t\t%d\t\t%d\t\t%d\n", i + 1, bt[i], wt[i], tat[i]);
}
printf("\nAverage Waiting Time: %.2f\n", total_wt / n);
printf("Average Turnaround Time: %.2f\n", total_tat / n);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo


divyank_rajput@divyankrajput: ~
divyank_rajput@divyankrajput:~$ nano round_robin.c
divyank_rajput@divyankrajput:~$ gcc round_robin.c -o round_robin
divyank_rajput@divyankrajput:~$ ./round_robin
Enter number of processes: 3
Enter burst time for each process:
P1: 24
P2: 3
P3: 3
Enter time quantum: 4
Process Burst Time Waiting Time Turnaround Time
P1 24 6 30
P2 3 4 7
P3 3 7 10
Average Waiting Time: 5.67
Average Turnaround Time: 15.67
divyank_rajput@divyankrajput:~$